



Department of Computer Science and Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E Computer Science and Engineering

Course Code & Title: CB3491 Cryptography and Cyber Security

Name of the Faculty member (s): Mrs.K.Jeyakarthika, AP/CSE

Date of Implementation : 10.10.2023

Innovative Practice Description

- **Unit / Topic:** Unit IV / MAC and HASH
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 12
- **Activity Chosen:** Flash cards
- **Justification:**

The topic chosen to perform this cross word puzzle activity is Hash Function, and MAC. Flash cards is the tool for the learners and the card hold the information. The topic MAC and Hash function is similar and the students need to learn the concepts. This activity helps the students to familiar with the concept in detail and also recall the concepts. This activity helps the students to understand the techniques and its complexity clearly.

- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation**
 - Instructor explained the particular concepts/topic in classroom within 35 minutes.
 - A flash card is a method in which the question and corresponding answer is in separate card.
 - Two types of Flash cards were prepared. Question and answer of the card is prepared based on topic taught by the course instructor in the class on a particular day. One containing question and other card containing answers.
 - Question cards were distributed to certain group of students and answer cards were distributed to few group of students.
 - Once the question will be asked by one of the students in the class from the card, student members of group in the class should identify correct answer cards and explain the answer in front of the class.
 - This will help the students to revise the particular topic.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO12
CO1	3	2	1	2	2	-	-	1	1	1

(1 – Low 2 – Moderate 3 – High)



PO / PSO mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO12
	3	2	1	2	2	1	1	1
	Basic Computer Science Engineering and mathematics Knowledge for implementing digital signature is highly correlated	Analyzing various digital signature schemes are highly correlated	Developing systems with basic digital signature mechanisms are highly correlated	Investigate various aspects of cryptanalysis is highly correlated	Use of virtual environment and crypt tool for performing digital signature is highly correlated	The students will adhere to the ethics.	The students will be able to work as an individual and team to how to distribute a key with integrity and authentication.	Updating the digital signature algorithms to the current trends are moderately correlated

• Images / Screenshot of the practice:

MAC

Message Authentication Code
Short Piece of Information Used to Authenticate a Message
Two Input (Original Message and a key)
Symmetric key
Examples: SHA1, SHA 2, SHA 3
Application: Financial Cryptography

HASH

Hash Based Message Authentication Code
It is called Hash or Digest with fixed size.
Single Input (Original Message)
Examples: HMAC, CMAC
Applications: Store Passwords, identify files



❖ Reflective Critique:

- This helped the students to identify the definition and also recalling the topic taught on that day.
- It motivates the students to identify the term associated with each definition and questions.
- Students said that the flash card activity is very joyful to recollect the concepts.
- They asked to conduct more of the similar activity during class which help us to listen the class without getting bored.

❖ Benefit of the practice:

- Based on the activity student's felt that it was easy to revise the concepts.
- Most of the students were actively participated and enjoyed among their group members and they answered correctly.

❖ Challenges faced in implementation:

- Few students found it difficult to understand the questions clearly and the instructor instructed the questioner to repeat the question once again clearly to make the students for clear understanding of question.
- Some students were hesitated to express the relevant answers to the question card.

References:

- ❖ <https://www.nyu.edu/faculty/teaching-and-learning-resources/strategies-for-teaching-with-tech/flipped-classes/implementing-a-flipped-class.html>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/2022-2023/CS8792-RV-Flipped-Unit-III.pdf>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2021-2022/Unit_4_flashcard.pdf

Signature of Faculty Member

HOD-CSE